

CTMT Collapse and Holonomy

Global Obstructions, Resolution Failure, and Observable Transport Groupoids

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Abstract

Chronotopic Metric Theory (CTMT) begins from the local observable resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma),$$

where O is the observable space, $R = \text{Im}(J)$ is the resolved sector induced by a local response operator, $N = \text{Ker}(J^T)$ is the unresolved sector, and Σ is empirical observation covariance. Previous work formulates admissible morphisms as transformations preserving this quadruple and shows that rank, Fisher-type response metrics, null covariance, covariance coupling, winding, and holonomy can be reconstructed as derived invariant structures.

The present paper develops the next foundational layer: collapse and holonomy. Collapse is defined operationally as failure of one or more levels of the resolution geometry closure hierarchy. Holonomy is defined as the groupoid obstruction obtained by composing admissible transition maps around closed chains of resolution charts. The main result is a logical lock: non-trivial holonomy is not necessarily collapse, but collapse is precisely the failure of bounded admissible trivialization of the transported resolution geometry. When rank, sector decomposition, covariance blocks, and holonomy remain inside admissible tolerances, CTMT predicts coherent observable transport; when any required component fails, the corresponding invariant functor must fail as well. This converts collapse from a metaphor into a falsifiable hierarchy of observable failures.

The paper gives theorem-level statements, proof sketches, counterexamples, non-claims, and a reproducible numerical protocol. The emphasis is not on claiming a physical mechanism, but on identifying the exact mathematical conditions under which observable transport geometry remains globally coherent or becomes operationally non-closed.

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1 Introduction

CTMT treats observable transport as a geometric reconstruction problem. The primitive object is not a metric, field equation, or hidden state space. It is the local resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma),$$

constructed from a local response operator J and empirical covariance Σ . The resolved sector $R = \text{Im}(J)$ contains first-order observable response directions, while $N = \text{Ker}(J^T)$ contains directions unresolved by the same local experiment. Covariance then induces the block decomposition

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

This paper asks what happens when such local geometries are transported through a family of charts. Two new phenomena become unavoidable.

- (i) *Holonomy*: composing locally admissible transitions around a loop may return a non-trivial automorphism of the initial resolution geometry.
- (ii) *Collapse*: a transported system may fail to preserve one or more components of the resolution geometry closure hierarchy.

The central point is that these notions are related but not identical. Holonomy is a global obstruction to trivialization. Collapse is a failure of admissible observable closure. A system may have non-trivial holonomy while remaining coherent if the holonomy lies inside the automorphism group of the resolution geometry. Conversely, collapse occurs when the transported object leaves the admissible class: rank changes, sector projectors fail, covariance blocks cease to match, null inflation becomes unbounded, or loop transport is not representable as an admissible automorphism.

2 Local Resolution Geometry Recalled

Definition 2.1 (Observable resolution geometry). A local CTMT resolution geometry is a quadruple

$$\mathcal{G} = (O; R, N, \Sigma),$$

where $O \cong \mathbb{R}^m$, $R = \text{Im}(J)$, $N = \text{Ker}(J^T)$, and Σ is a symmetric positive semi-definite empirical covariance operator on O .

Definition 2.2 (Resolved-null projectors). Let U_R and U_N be orthonormal bases for R and N . The corresponding orthogonal projectors are

$$P_R = U_R U_R^T, \quad P_N = U_N U_N^T.$$

They satisfy

$$P_R^2 = P_R, \quad P_N^2 = P_N, \quad P_R P_N = 0, \quad P_R + P_N = I.$$

Definition 2.3 (Admissible local automorphism). A linearized observable transformation $A : O \rightarrow O$ is locally admissible for $\mathcal{G}$ if

$$A(R) = R, \quad A(N) = N, \quad A^T \Sigma A = \Sigma.$$

The corresponding automorphism group is denoted $\text{Aut}(\mathcal{G})$.

Remark 2.4. The linear formulation is the tangent-level version of the smooth definition $\phi^* \mathcal{G} = \mathcal{G}$. It is sufficient for the collapse and holonomy criteria used in numerical batteries.

3 Closure Hierarchy

Collapse can only be meaningful after specifying what must remain closed. CTMT therefore uses a hierarchy rather than a single yes/no condition.

Definition 3.1 (Closure hierarchy). Let $\mathcal{G} = (O; R, N, \Sigma)$ be a resolution geometry and let A be a transport map. The closure hierarchy consists of the following levels:

- C1. Rank closure:** $\text{rank}(AJ) = \text{rank}(J)$.
- C2. Resolved closure:** $AP_R A^{-1}$ defines the transported resolved projector with unchanged dimension.
- C3. Null closure:** $AP_N A^{-1}$ defines the transported null projector with unchanged dimension.
- C4. Sector closure:** $A(R) = R$ and $A(N) = N$.
- C5. Covariance closure:** $A^T \Sigma A = \Sigma$ or, between charts, $A^T \Sigma_2 A = \Sigma_1$.
- C6. Coupling closure:** the block C_{RN} is preserved under the adapted resolved-null decomposition.
- C7. Information closure:** the induced response metric $F = J^T \Sigma^{-1} J$ is preserved up to admissible pullback.
- C8. Holonomy closure:** every closed chart loop produces an element of $\text{Aut}(\mathcal{G})$ with bounded condition number and admissible covariance error.

Proposition 3.2 (Nested closure). *Full admissibility implies every lower closure level. In particular, sector and covariance closure imply rank, resolved, null, coupling, and information closure whenever Σ is invertible on the resolved sector.*

Proof. Sector preservation fixes R and N , hence their dimensions and projectors. Covariance preservation fixes the covariance block decomposition. Since $F = J^T \Sigma^{-1} J$ is functorially constructed from J and Σ , it is preserved under admissible pullback. Rank closure follows from preservation of $R = \text{Im}(J)$. $\square$

4 Collapse

Definition 4.1 (Collapse event). A collapse event is a failure of at least one required closure level in a specified CTMT protocol. Equivalently, for a chosen tolerance vector

$$\varepsilon = (\varepsilon_{\text{rank}}, \varepsilon_R, \varepsilon_N, \varepsilon_\Sigma, \varepsilon_H),$$

a transported geometry collapses if one of the measured closure defects exceeds its admissible tolerance.

Definition 4.2 (Closure defects). For a candidate transport map A , define

$$\delta_{\text{rank}}(A) = |\text{rank}(AJ) - \text{rank}(J)|, \quad (1)$$

$$\delta_R(A) = \|AP_R A^{-1} - P_R\|, \quad (2)$$

$$\delta_N(A) = \|AP_N A^{-1} - P_N\|, \quad (3)$$

$$\delta_\Sigma(A) = \|A^T \Sigma A - \Sigma\|, \quad (4)$$

$$\delta_{RN}(A) = \|C_{RN}(A) - C_{RN}\|, \quad (5)$$

$$\delta_F(A) = \|F(A) - F\|. \quad (6)$$

A collapse is detected whenever any required defect is nonzero beyond numerical or operational tolerance.

Principle 4.3 (Falsification-first collapse principle). *CTMT does not infer collapse from visual instability, interpretive breakdown, or physical intuition. Collapse is declared only when an explicitly stated observable closure condition fails.*

Theorem 4.4 (Collapse as non-admissibility). *Let $A : O \rightarrow O$ be an invertible linearized transport map. If*

$$A \notin \text{Aut}(\mathcal{G}),$$

then at least one of sector closure or covariance closure fails. Consequently, A is a collapse map for any protocol requiring full admissibility.

Proof. By definition, $A \in \text{Aut}(\mathcal{G})$ exactly when it preserves R , N , and Σ . Therefore, if A is not an automorphism, at least one defining condition fails. These defining conditions are precisely sector and covariance closure. Any full-admissibility protocol therefore rejects A as collapsed. $\square$

Corollary 4.5 (Partial survival). *A transport map may preserve rank while still being collapsed at higher levels of the hierarchy.*

Proof. Rank is only the dimension of R . A map may preserve dimension while rotating resolved directions into null directions, changing covariance blocks, inflating null covariance, or breaking holonomy closure. Therefore rank preservation is necessary but not sufficient for CTMT admissibility. $\square$

5 Transport Groupoid and Holonomy

Definition 5.1 (Resolution chart). A resolution chart is a pair $(U_\alpha, \mathcal{G}_\alpha)$ where

$$\mathcal{G}_\alpha = (O_\alpha; R_\alpha, N_\alpha, \Sigma_\alpha)$$

is a local resolution geometry reconstructed on the observable neighborhood U_α .

Definition 5.2 (Admissible transition). A transition map

$$\phi_{\alpha\beta} : O_\beta \rightarrow O_\alpha$$

is admissible when

$$\phi_{\alpha\beta}^* \mathcal{G}_\alpha = \mathcal{G}_\beta.$$

At tangent level this means

$$D\phi_{\alpha\beta}^T \Sigma_\alpha D\phi_{\alpha\beta} = \Sigma_\beta, \quad D\phi_{\alpha\beta}(R_\beta) = R_\alpha, \quad D\phi_{\alpha\beta}(N_\beta) = N_\alpha.$$

Definition 5.3 (Resolution transport groupoid). The CTMT transport groupoid $\mathcal{T}(\mathcal{G})$ has resolution charts as objects and admissible transitions as morphisms. Composition is ordinary composition on overlaps; inverses exist whenever transitions are invertible.

Definition 5.4 (Holonomy). For a closed chain

$$\gamma = (\alpha_0, \alpha_1, \dots, \alpha_k, \alpha_0),$$

the holonomy is

$$H_\gamma = \phi_{\alpha_0\alpha_k} \circ \phi_{\alpha_k\alpha_{k-1}} \circ \dots \circ \phi_{\alpha_1\alpha_0} \in \text{Aut}(\mathcal{G}_{\alpha_0})$$

whenever every transition is admissible. The set of all such loop products is $\text{Hol}(\mathcal{G}_{\alpha_0}) \subseteq \text{Aut}(\mathcal{G}_{\alpha_0})$.

Theorem 5.5 (Holonomy is not collapse). *If all transitions along a closed chain are admissible, then the resulting holonomy belongs to $\text{Aut}(\mathcal{G}_{\alpha_0})$. Therefore non-trivial holonomy alone does not imply collapse.*

Proof. Admissible transitions are closed under composition in the transport groupoid. A closed chain begins and ends at the same chart, so its composite is an endomorphism preserving $\mathcal{G}_{\alpha_0}$. Since each transition is invertible, the endomorphism is an automorphism. It may be non-identity, but it still preserves the local resolution geometry. $\square$

Remark 5.6. This distinction is essential. Holonomy measures global obstruction to trivialization. Collapse measures failure of admissible closure. A curved but coherent geometry may have holonomy; a collapsed geometry fails admissibility.

6 The Collapse–Holonomy Lock

Definition 6.1 (Bounded admissible trivialization). A family of chart transitions admits bounded admissible trivialization if there exists a choice of local frames in which all transition maps and all loop holonomies satisfy

$$\delta_R, \delta_N, \delta_\Sigma, \delta_{RN}, \delta_F \leq \varepsilon$$

and

$$\kappa(H_\gamma) \leq K$$

for every tested loop γ , where K is a prescribed conditioning bound.

Theorem 6.2 (Collapse–holonomy lock). *Let $\{(U_\alpha, \mathcal{G}_\alpha)\}$ be a finite family of resolution charts with invertible transition candidates. Suppose the protocol requires sector, covariance, coupling, information, and holonomy closure. Then exactly one of the following occurs:*

- (a) *the chart family admits bounded admissible trivialization, and every closed loop produces admissible holonomy;*

(b) at least one transition or loop product violates a required closure level, and the family undergoes CTMT collapse relative to the protocol.

Proof. If bounded admissible trivialization exists, the stated closure defects are all within tolerance and loop products remain inside the admissible automorphism class with bounded conditioning. No collapse is detected. Conversely, if such trivialization does not exist, then by definition some transition or loop product exceeds a required defect tolerance or conditioning bound. This is exactly the operational definition of collapse. The alternatives are mutually exclusive by construction. $\square$

Corollary 6.3 (Logical lock). *Under an explicitly specified closure protocol, collapse is neither subjective nor interpretive. It is equivalent to failure of bounded admissible trivialization.*

7 Covariance Coupling and Automorphism Shrinkage

Definition 7.1 (Coupling amplitude). The coupling amplitude of a resolution geometry is

$$\chi_{RN} = \|C_{RN}\|_F.$$

Proposition 7.2 (Coupling constrains automorphisms). *If $C_{RN} = 0$, admissible automorphisms may act independently on resolved and null covariance blocks. If $C_{RN} \neq 0$, admissible automorphisms must also stabilize the cross-sector coupling block. Hence nonzero coupling weakens the available stabilizer freedom.*

Proof. When $C_{RN} = 0$, covariance is block diagonal relative to $O = R \oplus N$ and block-preserving transformations act independently subject to preserving Σ_R and Σ_N . When $C_{RN} \neq 0$, the equation $A^T \Sigma A = \Sigma$ includes off-diagonal constraints. These constraints couple the resolved and null actions and remove transformations that would have been admissible in the block-diagonal case. $\square$

Corollary 7.3 (Coupling-induced collapse risk). *Increasing $\|C_{RN}\|_F$ does not by itself imply collapse. It increases the rigidity of admissibility and therefore makes generic perturbations more likely to violate covariance or coupling closure.*

8 Counterexamples

Example 8.1 (Rank survives but geometry collapses). Let

$$A = \begin{pmatrix} I_R & B \\ 0 & I_N \end{pmatrix}, \quad B \neq 0.$$

For suitable B , $\text{rank}(AJ) = \text{rank}(J)$ may remain unchanged. However, $A(N) \not\subseteq N$ and the resolved-null decomposition is not preserved. Thus rank closure survives but sector closure fails.

Example 8.2 (Null inflation). Let

$$A = \begin{pmatrix} I_R & 0 \\ 0 & \lambda I_N \end{pmatrix}, \quad \lambda \neq 1.$$

Rank and sector dimensions remain unchanged, but

$$A^T \Sigma A \neq \Sigma$$

unless the null covariance is degenerate in a compatible way. Thus null inflation is a covariance collapse even when rank survives.

Example 8.3 (Non-trivial holonomy without collapse). Let each transition map preserve R , N , and Σ , but let the loop product be a non-identity rotation inside a covariance-degenerate resolved plane. Then $H_\gamma \neq I$, but $H_\gamma \in \text{Aut}(\mathcal{G})$. This is holonomy, not collapse.

Example 8.4 (Holonomy collapse). If the loop product satisfies

$$H_\gamma^T \Sigma H_\gamma \neq \Sigma$$

or has unbounded condition number relative to the protocol, then the loop is not an admissible automorphism. This is holonomy-level collapse.

9 Numerical Falsification Protocol

A reproducible CTMT collapse–holonomy battery should execute the following steps from one dataset.

Step 1. Construct the local response operator J by finite differences or local regression.

Step 2. Compute the SVD of J and determine R , N , P_R , P_N , and rank.

Step 3. Estimate empirical covariance $\Sigma = \frac{1}{n-1} R_{\text{obs}}^T R_{\text{obs}}$ from repeated residual observations.

Step 4. Add an explicitly stated ridge if needed:

$$\Sigma_\lambda = \Sigma + \lambda I.$$

Step 5. Lift covariance into the adapted basis $Q = [U_R \ U_N]$:

$$Q^T \Sigma_\lambda Q = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

Step 6. Compute Fisher-type response metric

$$F = J^T \Sigma_\lambda^{-1} J.$$

Step 7. Build transition candidates: identity, admissible sector rotations, resolved–null mixing, rank collapse, null inflation, covariance anisotropy, and loop products.

Step 8. For each candidate compute all closure defects.

Step 9. Declare pass/fail only from the defect table.

10 Master Diagnostic Table

Table 1: Collapse and holonomy diagnostics from one observable dataset.

Diagnostic	Construction	Collapse meaning
Rank	SVD of J	loss of resolved dimension
Resolved basis	U_R	drift of observable response sector
Null basis	U_N	drift of unresolved sector
Projectors	P_R, P_N	failure of sector decomposition
Covariance	Σ	observation geometry not preserved
Coupling	C_{RN}	resolved–null covariance relation changes
Fisher metric	$J^T \Sigma^{-1} J$	distinguishability geometry changes
Holonomy	loop product H_γ	global obstruction or loop collapse
Conditioning	$\kappa(H_\gamma)$	unbounded transport amplification

11 Non-Claims and Limitations

1. This paper does not claim that CTMT collapse is a physical singularity. It is an operational failure of observable resolution closure.
2. This paper does not claim that holonomy is itself collapse. Holonomy is allowed when it remains admissible.
3. This paper does not claim that rank preservation is sufficient. Rank is only the first health bar.
4. This paper does not claim that covariance coupling has a unique physical interpretation. It is treated as an observable cross-sector covariance block.
5. This paper does not solve globalization for arbitrary continuous chart families. It gives a finite chart/groupoid criterion suitable for computation and falsification.
6. This paper does not replace established differential geometry, information geometry, or transport theory. It formulates an observable resolution layer from which derived structures can be tested.

12 Open Problems

Open Problem I: Continuous holonomy classification. Classify $\text{Hol}(\mathcal{G})$ for smooth families of resolution geometries with nonconstant rank and covariance.

Open Problem II: Coupling-scale law. Determine whether $\|C_{RN}\|_F$ controls automorphism shrinkage by a sharp quantitative bound.

Open Problem III: Collapse thresholds. Develop statistically justified tolerance selection for projector, covariance, Fisher, and holonomy defects under finite-sample noise.

Open Problem IV: Physical realization. Identify physical transport systems for which covariance-induced coupling has a stable interpretable mechanism rather than being only an observational feature.

Open Problem V: Groupoid completion. Determine when a finite admissible chart groupoid extends to a smooth global resolution bundle.

13 Conclusion

Collapse and holonomy are the first genuinely global CTMT structures. Local resolution geometry identifies what is observable, unresolved, and statistically coupled. Holonomy asks whether those local geometries can be transported around loops without becoming globally trivial. Collapse asks whether the transported geometry remains admissible at all.

The resulting lock is precise. Non-trivial holonomy may survive. Rank alone may survive. But full CTMT coherence requires bounded admissible trivialization across the requested closure hierarchy. When this fails, the failure is not rhetorical: it appears as a measurable defect in rank, sector projectors, covariance blocks, Fisher geometry, coupling, or holonomy. This is the operational content of CTMT collapse.

A Appendix A: Minimal Python Battery Specification

A numerical implementation should emit at least the following JSON fields:

```
{
  "J": [...],
  "singular_values": [...],
  "rank": 4,
  "U_R": [...],
  "U_N": [...],
  "P_R": [...],
  "P_N": [...],
  "Sigma": [...],
  "Sigma_block": [...],
  "Sigma_R": [...],
  "C_RN": [...],
  "Sigma_N": [...],
  "Fisher": [...],
  "holonomy_products": [...],
  "closure_defects": [...],
  "pass_fail_table": [...]
}
```

B Appendix B: Recommended Pass/Fail Table

Table 2: Recommended collapse battery controls.

Control	Expected result	Reason
Identity	pass	preserves all structures
Sector rotation with compatible covariance	pass	admissible automorphism
Resolved-null mixing	fail	breaks sector closure
Rank collapse	fail	breaks rank closure
Null inflation	fail	breaks covariance closure
Covariance anisotropy	fail	breaks block covariance
Admissible loop product	pass	holonomy but not collapse
Non-admissible loop product	fail	holonomy-level collapse

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